

Independent reanalysis of the largest sporadic Creutzfeldt–Jakob disease genome-wide association study confirms all published loci and refines the STX6 signal

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Independent data-organization initiative. This work uses only publicly deposited, anonymized datasets; it involves no patient-identifiable data and makes no clinical claim.

Abstract

Background. Sporadic Creutzfeldt–Jakob disease (sCJD) is the most common human prion disease. The largest sCJD genome-wide association study (GWAS) identified PRNP, STX6 and GAL3ST1 at genome-wide significance in 4,110 cases and 13,569 controls; its summary statistics are public and have since been used by other groups for multi-omic risk-gene prioritization and for colocalization method development. We report an audit conducted independently of those analyses.

Methods. We reanalysed the deposited summary statistics (GCST90001389; 6,314,492 variants, GRCh37) end to end, implementing all inferential statistics from first principles and cross-validating them against R: genomic-control inflation, genome-wide significance screening, Wakefield approximate Bayes factors, linkage-disequilibrium clustering against 1000 Genomes phase 3, and stratified inflation by minor-allele-frequency bin. We additionally reanalysed two companion expression datasets (GSE160208 brain; GSE140069 whole blood).

Results. We independently reproduced **all three published loci** — PRNP chr20:4,672,307 ($p = 1.62 \times 10^{-15}$), GAL3ST1 chr22:30,950,360 ($p = 6.18 \times 10^{-10}$), STX6 chr1:180,961,245 ($p = 7.51 \times 10^{-9}$) — recovered the published STX6 index variant rs3747957 at its exact reported value ($p = 9.74 \times 10^{-9}$, $\beta = -0.148$), and found no malformed records; global $\lambda = 1.0587$, varying by only 0.0156 across allele-frequency strata. Fine-mapping shows the STX6 signal is one cohesive haplotype block carrying 90.5% of regional posterior mass. A blind clumping scan rediscovered the three known loci without locus hints (3/3), indicating pipeline sensitivity rather than circular confirmation. A five-cohort brain eQTL meta-analysis (~2,182 samples) renders the STX6 lead variant a genome-wide significant eQTL ($p = 7.60 \times 10^{-47}$) with 89% direction concordance, reproducing the expression–disease relationship previously established with a larger multi-omic design. No significant cis-sQTL was detected for STX6 in any cohort; because only significance-passed pairs are available, this bounds what we can detect rather than establishing absence. In GSE140069, the blood microRNA signature falls from 84 FDR-significant miRNAs unadjusted to 1 after adjustment for age, sex and RNA integrity; isolating each covariate attributes ~97% of that collapse to age (all variance-inflation factors < 1.4).

Conclusions. Public prion-disease summary statistics support rigorous independent verification by non-consortium parties. This study confirms the published loci and their local LD structure, quantifies the ceiling of summary-statistics-only fine-mapping, and measures how much covariate confounding inflates a blood-based biomarker signature. All code and reports are openly available.

1. Introduction

Human prion diseases are fatal neurodegenerative conditions caused by misfolding of the cellular prion protein (PrP^{Sc}). Sporadic CJD (sCJD) accounts for ~85% of cases. Beyond the PRNP codon 129 modifier, host genetic modifiers were established by the largest sCJD genome-wide association study to date [1], which identified PRNP, STX6 and GAL3ST1 at genome-wide significance in 4,110 cases and 13,569 controls.

That study's summary statistics are publicly deposited (GCST90001389), and they have already been analysed by other groups: the multi-omic risk-gene study [4] integrated them with eQTL, pQTL, SMR and colocalization, and the deposited statistics have served as test data for colocalization method development [8]. We make no novelty claim for accessing these data. What remains useful is an audit conducted *independently of both* — one

that verifies the deposition on its own terms, stress-tests the published conclusions with a separately implemented statistical core, and quantifies — with numbers a reader can reproduce — how much of the reported signal survives standard covariate adjustment in the companion biomarker datasets.

Independent reanalysis serves three purposes the original analysis cannot: (i) it verifies that public deposition is complete and internally consistent; (ii) it stress-tests conclusions with methods chosen independently of the original pipeline; (iii) it lowers the barrier for citizen-science participation in rare-disease research, following the precedent of Minikel & Vallabh [10].

Here we report a fully independent, from-scratch reanalysis of GCST90001389 and two companion expression datasets (GSE160208 brain tissue; GSE140069 whole blood), executed without access to individual-level genotypes and with all inferential statistics implemented from first principles. Where our results reproduce published findings we say so and attribute them; where they add something, we state precisely what is new and what is not.

2. Data

Resource	Accession	Content
sCJD GWAS sumstats	GCST90001389	6,314,492 variants, GRCh37, β /SE/p/EAF
Brain expression	GSE160208	47 samples (27 CJD / 20 controls), NanoString 800 genes
Blood microRNA	GSE140069	57 sCJD / 48 controls, 939 miRNAs
LD reference	1000 Genomes phase 3 (ALL)	via Ensembl REST ld endpoint

Integrity: SHA-style MD5 checksums recorded before analysis (Appendix of project repository).

3. Methods

3.1 GWAS quality control

Streaming parse of the compressed sumstats (single pass, constant memory). Malformed-line count, allele-frequency sanity, χ^2 statistic per variant as $(\beta/\text{SE})^2$, genomic-control factor $\lambda = \text{median}(\chi^2)/0.454936$ computed globally and within pre-registered MAF strata (<0.05 ; $0.05\text{--}0.25$; $0.25\text{--}0.45$; ≥ 0.45). Genome-wide threshold $p < 5 \times 10^{-8}$.

3.2 Locus annotation

GRCh37 gene coordinates retrieved live from Ensembl REST (never from memory) for PRNP, STX6, GAL3ST1 windows (± 50 kb around lead).

3.3 Approximate-Bayes-factor fine-mapping

Per-variant Wakefield ABF with prior variance $W = 0.04$ on $\log(\text{OR})$: $\text{ABF}_i = \sqrt{(\text{SE}^2/(\text{SE}^2+W)) \cdot \exp(\chi^2_{iW}/(2(\text{SE}^2+W)))}$. Regional posteriors $\pi_i \propto \text{ABF}_i$. Because individual genotypes are unavailable, we report **cluster-level credible mass**: variants are grouped by pairwise $r^2 \geq 0.80$ with the regional lead using Ensembl REST LD (1000G phase 3 ALL); posterior mass of the lead cluster versus remainder quantifies whether the signal is one haplotype block or dispersed. This is explicitly an approximation — joint models (SuSiE/FINEMAP) require genotypes.

3.4 Expression analyses

GSE160208: Welch t-tests on $\log_2$ -transformed normalized intensities, frontal-cortex-only contrasts, Benjamini–Hochberg FDR; exact replication of the original authors' criteria ($p < 0.05 \wedge |\log_2 \text{FC}| > 1$) alongside our

FDR<0.05 criterion. GSE140069: log2(x+1) transform; primary model = OLS log2 ~ group + sex + age + RIN (covariates from the series matrix; the original publication adjusted age); sensitivity analyses unadjusted and detection-filtered. Effect sizes as Cohen's d. All statistics cross-validated against R anchors (t distribution CDF and BH procedure matched to $\leq 10^{-13}$ relative error; permutation calibration of FDR).

3.5 Colocalization with brain eQTLs

For the STX6 region (chr1:180.9–181.1 Mb GRCh37), we tested whether the GWAS signal and STX6 expression quantitative trait loci (eQTLs) share a causal variant, using the regional approximate-Bayes-factor colocalization framework (Giambartolomei et al. 2014) implemented from the published equations: prior $W = 0.04^2$; priors $p_1 = p_2 = 10^{-4}$, $p_{12} = 10^{-5}$ (sensitivity: 10× more conservative). eQTL summary statistics were retrieved as remote tabix region queries (pysam/htslib) from the eQTL Catalogue release 8, which uniformly reprocessed GTEx v10 and independent brain cohorts: CommonMind DLPFC (n=586), ROSMAP DLPFC (n=560), BrainSeq DLPFC (n=479), GTEx v10 DLPFC (n=285) and GTEx v10 cerebellum (n=272). eQTL coordinates (GRCh38) were converted to GRCh37 by subtracting a single constant offset (–30,864 bp) determined from the STX6 gene boundaries. **We verified this offset against the Ensembl assembly-mapping service**, sampling points across the full STX6 span in GRCh37: the returned GRCh38 coordinate is exactly +30,864 bp at every point sampled, and every eQTL variant that pairs with a GWAS variant under the constant also pairs under the service's own mapping. A constant offset is exact only where no assembly-level indel intervenes, so this verification bounds rather than eliminates the concern — a reimplementa-tion should still prefer a chain-file liftOver — but for the STX6 window it holds empirically, and the colocalization result does not depend on the approximation. Alleles were harmonised to the ALT/effect convention (0 incompatible variants discarded). Because no single cohort is individually powered, we additionally performed an inverse-variance-weighted meta-eQTL across the five cohorts (~2,182 brain samples), yielding a meta-eQTL signal at the GWAS lead variant of $p = 7.60 \times 10^{-47}$ (the regional minimum, a different variant, reaches $p = 6.60 \times 10^{-47}$). Implementation validation: (i) an independent reimplementa-tion of the ABF/posterior equations in R agrees with our Python code to 6 decimal places ($H_3 = 0.994997$ vs 0.9950); (ii) the standard R `coloc.abf` (coloc 5.2.3) reproduces the combined both-signal posterior ($H_2 + H_3 + H_4 = 1.000$) with per-variant PP.H4 Spearman $\rho = 0.91$ against our ABF-diagonal mass — while demonstrating that the shared-vs-distinct split (H_4 vs H_3/H_2) is prior-dependent, as declared. LD-panel sensitivity: re-running the fine-mapping with the population-matched 1000 Genomes EUR panel leaves the STX6 lead-cluster posterior unchanged (90.5% at $r^2 \geq 0.8$), improves PRNP tagging (58.9% → 80.9%), and leaves GAL3ST1 poorly tagged in both panels (consistent with a lower-frequency haplotype). Splicing scan: leafcutter sQTL summary statistics for the same five cohorts (eQTL Catalogue r8 cc files = significance-passed cis pairs per intron cluster; cluster identity normalised by genomic coordinates) were harmonised to the GWAS and tested by colocalization per cohort×cluster; clusters were attributed to genes by the catalogue's `gene_id` column (STX6 = ENSG00000135823, GRCh37 chr1:180,941,861–180,992,047, verified via Ensembl REST lookup).

3.6 Blind whole-genome clumping (discovery sensitivity)

To demonstrate that our pipeline would find the known loci rather than merely confirm them, we scanned all 6,314,492 variants blind (threshold $p < 10^{-5}$) and applied greedy distance clumping (± 500 kb; lead = smallest p within cluster), with no locus annotation used until after cluster definition. Known loci were matched post hoc.

4. Results

4.1 Sumstats integrity and inflation

Zero malformed records among 6,314,492 lines. Global $\lambda_{GC} = 1.0587$ (liminal, acceptable). Stratified λ :

MAF stratum	n	λ
<0.05	325,236	1.0617
0.05–0.25	532,664	1.0579
0.25–0.45	328,920	1.0547
≥ 0.45	76,078	1.0703

Gradient 0.0156 — inconsistent with major residual population stratification, which preferentially inflates common variants.

4.2 Independent replication of all three loci

Locus	Our best hit (GRCh37)	p	β	Annotation
PRNP	chr20:4,672,307 C>T	1.62×10^{-15}	-0.219	PRNP region; rs60704301/rs2093390/rs4254562
GAL3ST1	chr22:30,950,360 T>C	6.18×10^{-10}	-0.169	GAL3ST1 promoter/5' region
STX6	chr1:180,961,245 G>A	7.51×10^{-9}	-0.149	intragenic STX6 (Ensembl 180,941,861-180,992,047)

All 41 genome-wide-significant variants fall within these three regions; none elsewhere. A blind whole-genome clumping scan (Section 3.6) independently rediscovers all three loci as the top three clusters, confirming pipeline sensitivity rather than circular confirmation.

4.3 Cluster-level fine-mapping

- **PRNP**: 337 regional variants; anchor rs60704301 (merged $\rightarrow$ rs2093390). Posterior mass 100% within $r^2 \geq 0.50$ of the anchor in **both LD panels** (58.9% at $r^2 \geq 0.80$ with the ALL panel; **80.9%** with the population-matched EUR panel); codon-129 variant rs1799990 appears among proxy rsIDs. The entire signal is one haplotype structure around the prion-protein gene. - **STX6**: 162 variants; anchor = lead rs11586493; 20/20 top variants panel-covered, max $r^2 = 1.00$; **90.5%** of mass in the $r^2 \geq 0.80$ lead cluster — a single cohesive block that includes rs3747957 ($r^2 = 0.99$); **identical (90.5%) under the EUR panel**. - **GAL3ST1**: 322 variants; lead absent from 1000G phase 3 (anchor rs386462923 $\rightarrow$ rs8142452, rank 3); only 4/20 top variants panel-covered; best cross- $r^2 \approx 0.45$ — **unchanged under the EUR panel**. The signal is poorly tagged by common proxies — consistent with a lower-frequency haplotype and an explicit caveat for imputation-based replication.

4.4 Independent recovery of the STX6 index variant from the deposited statistics

The index variant at STX6 is **rs3747957** (chr1:180,953,853 GRCh37). This variant was reported by the original consortium [1] — it appears in the abstract of that paper, in Table 1, in the independent replication cohort (1,098 cases / 498,016 controls), in their eCAVIAR fine-mapping and PAINTOR functional annotation, and was linked there to a GTEx putamen eQTL ($p = 2.3 \times 10^{-13}$). It was subsequently carried forward as the STX6 anchor by an independent multi-omic study [4]. **Neither of those findings is claimed as novel here.**

What this reanalysis contributes is an *independent reconstruction from the deposited summary statistics alone*. Recovering rs3747957 directly from the sumstats — with no access to individual genotypes and no reference to either source paper's tables — returns $p = 9.74 \times 10^{-9}$ and $\beta = -0.148$, matching the published values to the reported precision, with the same effect direction as our lead variant ($\beta = -0.149$); it ranks 11th of 162 regional variants by p-value. The value of this recovery is not the association itself but the LD architecture it permits us to quantify: rs3747957 sits at $r^2 = 0.99$ from our lead, inside a single cluster carrying 90.5% of regional posterior mass (§4.3). Reconstructing that block structure from deposited statistics — rather than from the original authors' fine-mapping output — is what allows an independent party to confirm that the STX6 signal is one cohesive haplotype rather than a set of independent false positives, and to quantify how much of the regional posterior mass any given analysis method leaves unattributed.

4.5 Expression signatures replicate exactly (brain) but are fragile under covariate adjustment (blood)

Brain GSE160208: 437/800 genes $FDR < 0.05$; 184 DEGs under original criteria — identical count and rank order ($r = 1.000$ top-10). Blood GSE140069: unadjusted log2-Welch yields 84 significant miRNAs ($10^{\uparrow 74} \downarrow$); OLS adjusting age+sex+RIN leaves **1** (hsa-miR-500a-3p); the four discovery miRNAs retain direction and nominal significance ($p = 7 \times 10^{-4} - 4 \times 10^{-2}$) but only hsa-miR-93-5p survives FDR within the detection-filtered universe ($q = 0.048$). Cases were on average 12.8 years older than controls (66.4 vs 53.6, Cohen's $d = 1.07$) with lower RNA integrity (RIN 5.59 vs 6.50, $d = -0.71$) — a textbook confounding structure that the original paper partially addressed (age via Partek GSA). We isolated each covariate rather than inferring the cause from the joint adjustment: age alone accounts for 84 of the 87 lost signals ($88 \rightarrow 4$ FDR-significant), RNA integrity alone for 62 ($88 \rightarrow 26$), and sex for 3 ($88 \rightarrow 85$); all variance-inflation factors are below 1.4, so these contributions are separately identifiable rather than collinear. Age is thus the dominant confounder and accounts for ~97% of the collapse, with RNA integrity accounting for the remainder. Age-restricted analyses within age-overlapping strata (50–70 years, residual gap +2.7 years) leave 5–7 miRNAs, but are underpowered at 22–29 controls per stratum and bound rather than resolve any residual effect. Furthermore, a novel cross-compartment integration analysis (not performed by either original study) shows that experimentally validated targets of the four discovery blood miRNAs are NOT over-represented among the up-regulated brain DEGs (hypergeometric, all $q = 1.0$; overlaps at or below chance), arguing that the blood signature reflects peripheral processes rather than the cerebral transcriptional program — cautioning against naive blood→brain causal inference.

4.7 Colocalization: the STX6 GWAS signal and a brain eQTL occupy the same region

Per-cohort colocalization is power-limited (eQTL $n \leq 586$; the GWAS lead is only a suggestive single-cohort eQTL, with concordant direction in 78–81% of regional variants). The five-cohort meta-eQTL is more informative: it renders the lead GWAS variant a **genome-wide significant STX6 eQTL** ($p = 7.60 \times 10^{-47}$; $z \approx 14$) with **89% effect-direction concordance** across 390 harmonised variants. (The strongest eQTL in the region is a different variant, at chr1:180,949,780, with $p = 6.60 \times 10^{-47}$; we report the lead variant's value throughout and name this one explicitly to avoid conflating the two.)

In the regional colocalization framework the posterior collapses onto the hypotheses in which both traits have a causal variant in the region: **$H_0 + H_1 \approx 0$ and $H_2 + H_3 + H_4 = 1.000$ in every parametrization tested** ($H_3 + H_4 \approx 0.995$ under our default prior; the standard R `coloc.abf` 5.2.3 with its own priors assigns 98% to H_4). This needs stating precisely, because the distinction matters: **H_2 and H_3 are hypotheses of two distinct causal variants**, not a null hypothesis. What the collapse establishes is that the GWAS signal and a brain eQTL are both genuinely associated within STX6 — it does not by itself establish that they are driven by the same variant. Only H_4 asserts a shared causal variant, and the split between H_4 and H_3 is formally unidentifiable here, because all cluster members are in $r^2 \geq 0.97$ LD. The 98% figure quoted from `coloc.abf` is therefore prior-dependent, not an estimate of shared causality: our own prior moves $H_3 + H_4$, and `coloc` assigns the mass to H_4 differently. We report the block-level posterior and the direction concordance rather than an H_4 point estimate.

Our implementation was validated by an exact reimplementations of the ABF/posterior equations in R (agreement to six decimal places) and against the standard `coloc.abf` package (per-variant PP.H4 Spearman $\rho = 0.91$ against our ABF-diagonal mass).

Taken together — a genome-wide significant meta-eQTL at the GWAS lead, 89% direction concordance, a single fine-mapping cluster, and the exclusion of H_0/H_1 — the evidence is consistent with, but does not establish, regulation of STX6 expression as the mechanism underlying the association. This conclusion reproduces work already reported independently [4], which reached the same interpretation using a larger multi-omic design including protein-level QTLs.

4.8 Splicing: no significant cis-sQTL for STX6 detected in any of the five brain cohorts

The eQTL Catalogue release 8 provides leafcutter splicing-QTL summary statistics for the same five brain cohorts. We scanned every significant intron-cluster whose cis window intersects the STX6 region, harmonised variants to the GWAS and ran the validated colocalization per cohort $\times$ cluster (6 tests in total). **No significant**

cis-sQTL cluster for STX6 was detected in any of the five cohorts (0/5), in contrast to the strong expression QTL at the same lead variant ($p = 7.60 \times 10^{-47}$); and no neighbouring-gene splicing signal (KIAA1614, QSOX1, RP5-1180C10.2) colocalizes with the GWAS either (all $PP.H4 \approx 0$, with a Bonferroni-aware reading).

The strength of this negative result is limited by the data available to it, and we do not claim otherwise. The cc files contain only significance-passed cis pairs: a cluster with no significance-passed pairs is **invisible to this scan**, so a true but weak or underpowered sQTL would produce the same output as a genuinely absent one. Absence of evidence in significance-filtered files is not evidence of absence. The defensible statement is therefore narrower than "splicing does not operate here": *within the significance-passed fraction of these five cohorts, we detected no splicing signal at STX6 to interpret*. A claim about the relative importance of expression versus splicing at this locus would require either unfiltered sQTL statistics or a cohort with adequate power for splicing effects.

4.9 Blind clumping rediscovers all published loci

The blind scan produced 35 independent clusters; the top three are exactly PRNP (chr20:4,672,307; $p = 1.62 \times 10^{-15}$), GAL3ST1 (chr22:30,950,360; $p = 6.18 \times 10^{-10}$) and STX6 (chr1:180,961,245; $p = 7.51 \times 10^{-9}$) — **3/3 known loci rediscovered without any locus hint**, in correct significance rank. The strongest residual cluster (chr16:15,539,902, near BMERB1; $p = 5.73 \times 10^{-8}$) sits just below the genome-wide threshold. **This is not a new locus**: the signal was recorded as subthreshold by the original consortium [1] and the same variant was functionally prioritized by the independent multi-omic analysis [4]. We re-observe it because an unbiased scan recovers it in the residual, which is a useful check on the analysis rather than a discovery; it still requires independent replication in a cohort with adequate power.

4.10 Ethics and scope

No individual-level human data were generated or obtained beyond public deposits; no patient-identifiable information was processed. This work is a verification contribution and makes no clinical claims.

5. Discussion

Our results deliver the three things an independent reanalysis can uniquely provide. First, **verification**: every number in the deposited sumstats parsed cleanly, all 41 genome-wide-significant variants sit exactly where the original consortium reported them, and genomic inflation is liminal and uniform across allele-frequency strata — the public record of the largest sCJD GWAS is trustworthy, and we publish the checksums and code to let anyone re-verify this in minutes.

Second, **an independent reproduction of the STX6 expression association using a different eQTL resource**. That genetically increased STX6 expression and protein abundance in brain raise risk of sCJD is established work [4], which integrated eQTL and pQTL data with SMR and colocalization across 43 authors; the same direction of association has also been reported for Alzheimer disease protein abundance [9]. We do not claim this as a discovery. What we add is a reconstruction from a different source: a five-cohort brain meta-eQTL built from eQTL Catalogue release 8 (~2,182 samples), which makes the GWAS lead variant a genome-wide significant STX6 eQTL ($p = 7.60 \times 10^{-47}$; the regional minimum, a different variant, reaches $p = 6.60 \times 10^{-47}$) with concordant effect direction in 89% of regional variants, and a regional colocalization that places the entire posterior on the both-signals hypotheses. Reproducing the association from an independent eQTL resource, with a different cohort set and a different significance threshold, is corroboration of the underlying signal rather than a new mechanistic claim. Strict single-variant attribution (H4) is unidentifiable under $r^2 \geq 0.97$ — we say so explicitly — and our posterior split is prior- dependent (§4.7).

Third, **an independent reconstruction of the STX6 haplotype block from deposited statistics alone**. Using only the summary statistics and no access to the original authors' fine-mapping output, we recover rs3747957 [1,4] at its published $p = 9.74 \times 10^{-9}$ and effect direction, and show it lies at $r^2 = 0.99$ from our own lead inside a single cluster carrying 90.5% of regional posterior mass (§4.3, §4.4). This is a verification result rather than a discovery: the association and the index variant were both already published [1,4]. What it establishes is that the deposited statistics are complete and internally consistent enough to reproduce a published locus and its local

LD structure from scratch — and it quantifies, for the first time in this locus, how much regional posterior mass a summary-statistics-only analysis can attribute (90.5%) versus how much remains unattributed, which is the honest ceiling of cluster-level inference.

Fourth, a **cautionary quantification for biomarker research**. The blood microRNA signature for sCJD (Nat Commun 2020) collapses from 84 FDR<0.05-significant miRNAs (no covariate adjustment) to 1 after standard covariate adjustment. Isolating each covariate shows why: age alone (cases 12.8 years older, $d = 1.07$) removes 84 of the 87 lost signals, RNA integrity alone removes 62, and sex removes 3, with all variance-inflation factors below 1.4 — so the collapse is attributable rather than merely incidental, and ~97% of it is age. Directionality and nominal significance of the four discovery miRNAs survive — consistent with a genuine but weaker-than-presented signal. We stress this is not an accusation: the original authors adjusted age themselves; our contribution is making the magnitude of naive-vs-adjusted divergence, and its attribution, explicit and reproducible for future biomarker pipelines.

Methodologically, we demonstrate that a complete GWAS quality-control, fine-mapping and colocalization pipeline can run with a standard-library statistical core (all inferential statistics implemented from first principles and cross-validated against R anchors), relying on external open-source libraries only for figure rendering (matplotlib) and remote tabix access to eQTL summary statistics (pysam/htslib). This lowers the entry barrier for researchers in resource-limited settings, including in countries like Brazil where sCJD surveillance exists but prion-genetics capacity is thin.

Finally, our cluster-level credible analysis shows the STX6 signal is one cohesive LD block whose posterior mass concentrates on the lead haplotype rather than dispersing across independent false positives. Formal joint fine-mapping with individual genotypes remains the gold standard; we offer ours as the honest ceiling achievable from summary statistics alone.

6. Limitations

- Summary-statistics-only: no conditional/joint fine-mapping; cluster-level inference only. - Expression datasets lack individual-level covariates for GSE160208 (no age/PMI metadata). - Our OLS implementation cannot reproduce Partek's gene-specific variance correction; differences in Section 4.5 may partly reflect estimator choice. - Colocalization under $r^2 \geq 0.97$ cannot split H3 (distinct variants) from H4 (shared); we report block-level support plus direction concordance instead of an H4 point estimate. - Brain eQTL cohorts (DLPFC/cerebellum, adult) only approximate the affected tissue and cell types in sCJD; cell-type-specific (e.g. microglial) eQTL may differ. - Single-author independent initiative; peer review pending (bioRxiv DOI upon submission).

7. Data availability

GWAS summary statistics. GCST90001389 (6,314,492 variants, GRCh37; GWAS Catalog, EMBL-EBI). **Expression datasets.** GSE160208 (brain, NanoString 800-gene panel) and GSE140069 (whole blood small-RNA-seq), both from the NCBI Gene Expression Omnibus. **LD reference.** 1000 Genomes Project Phase 3, accessed via the Ensembl REST 1d endpoint.

8. Code availability

All analysis code, reports containing every intermediate number, figure-generation scripts and input checksums are version-controlled at <https://github.com/BlackYuriJDU/sCJD-GWAS-reanalysis> (release v1.0.0-preprint-clean; archived on Zenodo). The inferential statistics — approximate Bayes factors, t and F distributions, Benjamini–Hochberg FDR, Welch's test, ordinary least squares — are implemented from first principles and cross-validated against R anchors (coloc.abf 5.2.3; see §4.7). External libraries are used only for figure rendering (matplotlib), tabular I/O (openpyxl) and remote tabix access to eQTL summary statistics (pysam/htslib). Software versions are pinned in requirements.txt; every figure and every reported number is regenerated from the committed scripts.

9. References

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Figures: volcano_gse160208.png, volcano_gse140069.png, heatmap_top_genes.png (pipeline/reports/figuras/). Fine-mapping numbers from relatorio_finemap_loci.md v2 and relatorio_lambda_gc.md. All inferential statistics cross-checked against R anchors.